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MILITARY SPECIFICATION

# SWITCHES, ROTARY, SHIELDED, AIRCRAFT IGNITION

*This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.*

## 1. SCOPE

1.1 **Scope.** This specification covers the general requirements for shielded ignition switches for use in aircraft ignition systems.

1.2 **Classification.** Swtiches shall be of the following types, as specified (see 6.2):

| *AN Part No.* | *Description* |
|---|---|
| AN3212-1 | Single engine |
| AN3213-1A | Two engine |
| AN3214-1A | Four engine |

## 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

### SPECIFICATIONS

FEDERAL

QQ-A-591 — Aluminum Alloy Die Castings

QQ-A-601 — Aluminum Alloy Sand Castings

QQ-P-416 — Plating, Cadmium (Electrodeposited)

VV-P-236 — Petrolatum, Technical

MILITARY

JAN-S-44 — Shock-Testing-Mechanism for Electrical-Indicating Instruments (2½ and 3½ Inch, Round, Flush-Mounting, Panel-Type)

MIL-E-5272 — Environmental Testing, Aeronautical and Associated Equipment, General Specification for

MIL-E-5557 — Enamel; Heat-Resisting, Glyceryl-Phthalate, Black

MIL-S-7742 — Screw Threads, Standard Optimum Selected Series: General Specification for

MIL-P-8585 — Primer Coating, Zinc Chromate, Low-Moisture-Sensitivity

MIL-A-8625 — Anodic Coatings, for Aluminum and Aluminum Alloys

FSC 5930

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MIL–E–17555 — Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery

MIL–E–25109 — Engines; Aircraft, Reciprocating, General Specification for

MIL–D–70327 — Drawings, Engineering and Associated Lists

STANDARDS

MILITARY

MIL–STD–130 — Identification Marking of U.S. Military Property

MIL–STD–143 — Specifications and Standards, Order of Precedence for the Selection of

MS33558 — Numerals and Letters, Aircraft Instrument Dial, Standard Form of

AIR FORCE-NAVY AERONAUTICAL

AN3212 — Switch — Single Engine Ignition

AN3213 — Switch—Two Engine Ignition

AN3214 — Switch—Four Engine Ignition

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

## 3. REQUIREMENTS

**3.1 Qualification.** The switches furnished under this specification shall be a product which has been inspected and has passed the qualification inspections specified herein, and has been listed on or approved for listing on the applicable Qualified Products List.

**3.2 Materials.**

3.2.1 *Nonmagnetic materials.* Unless otherwise specified herein, nonmagnetic materials shall be used wherever practicable.

3.2.2 *Fungus proofing.* Materials which are not nutrients for fungi shall be used to the greatest extent practicable. Where materials that are nutrients for fungi must be used, such materials shall be treated with a fungicidal agent, as approved by the procuring activity.

3.2.3 *Protective treatment.* When materials are used in the construction of the switches that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance of the switches as specified in this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.2.4 *Selection of materials.* Specifications and standards for all materials and parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be used in accordance with procedures established by the procuring activity, except as provided in the following paragraph. The order of precedence for the selection of specifications and standards shall be in accordance with Standard MIL–STD–143.

3.2.4.1 *Standard parts.* Military standard parts (MS, AN, and JAN) shall be used wherever they are suitable for the purpose,



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and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, and cotter pins, may be used, provided they possess suitable properties and are replaceable by the military standard parts without alteration, and provided the corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to this specification.

**3.3 Design and construction.**

**3.3.1** *Design.* The design of ignition switches shall conform to the applicable AN standard. Ignition switches shall provide selective "ON" and "OFF" electrical control of the complete ignition system of aircraft. Ignition switches and all connections thereto shall be completely and effectively shielded to eliminate interference noises in the aircraft receiving equipment. Switches shall be provided with at least one external connection suitable for grounding the conduit boss and the switch case to the airplane structure.

**3.3.2** *Construction.* The ignition switch shall consist of a faceplate, switches, a mounting flange, and operating mechanism enclosed in a case or cover, and a conduit boss. The switch shall be built to withstand the strains, jars, vibrations, and such other conditions as are incident to shipping, installation, service, and storage. Switches shall conform to the dimensions shown on the applicable AN standard.

**3.3.2.1** *Faceplates.* Faceplates shall be made of aluminum if they are separate from the mounting flange and are not subject to stress.

**3.3.3** *Switches.*

**3.3.3.1** *Individual engine switches.* Individual engine switches shall operate by a rotating shaft actuated by a "wing" handle, which shall be made of aluminum alloy conforming to Specification QQ–A–591, and shall have a durable dull finish which contrasts with the faceplate. The handle shall be of such design and shape as to be readily operable by a gloved hand. These handles shall not extend beyond the circle through the centers of the holes in the mounting flanges. Each switch shall function in any of four operation positions which shall control electrical connections for four circuit conditions as follows:

| Switch position marked on face of switch | Circuit condition |
|---|---|
| "OFF" | Both main magneto circuits closed (grounded) and not operating. |
| "R" | Right main magneto circuit open and operating. Left main magneto circuit closed (grounded) and not operating. |
| "L" | Left main magneto circuit open and operating. Right main magneto circuit closed (grounded) and not operating. |
| "BOTH" | Both main magneto circuits open and operating. |

All individual engine switch mechanisms shall be identical with respect to design, operation, and circuit condition control. The switch position order, "OFF," "R," "L," "BOTH," shall apply for clockwise rotation of the handles (facing switch). The individual engine switch handles shall be provided with positive stops to prevent rotation beyond the "OFF" or "BOTH" positions. Suitable means shall be provided for securely positioning the switch in any of the four positions without the necessity of additional



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manual operation. "Bottoming" in any of the four positions shall be positive and self-indicating. The total angular rotation of the switch handles shall not exceed 120 degrees.

3.3.3.2 *Single engine switches.* Single engine ignition switches shall be in accordance with Standard AN3212.

3.3.3.3 *Master switch.* The master (nonselective) switch shall be incorporated in switches AN3213–1A and AN3214–1A only. This switch shall be of the push-pull plunger type, and shall be operated by a flat circular knob located at the center of the switch faceplate. The handle of this switch shall be made of aluminum alloy conforming to Specification QQ–A–591, or of a type acceptable to the Government, and shall have a durable dull finish which contrasts with the faceplate. The plunger rod shall be of nonmagnetic, corrosion-resisting steel or other material acceptable to the Government. The design, shape, and position of the knob relative to the individual engine switch handles shall be such as to permit it to be quickly grasped by a gloved hand and pulled to the "OUT" (off) position. The total axial movement of the knob from "IN" to "OUT" shall not exceed ½ inch. A positive snap stop device shall be provided for securely positioning the master switch plunger in either the "IN' or "OUT" position without the necessity of additional manual operation. "Bottoming" in either the "IN" or "OUT" position shall be positive and self-indicating. The two positions of the master switch plunger shall control circuit conditions as follows:

| Position | Circuit condition |
|---|---|
| "IN" | All main magneto circuits open and subject to control by individual engine switches. |
| "OUT" | All main magneto circuits closed (grounded). |

3.3.4 *Mounting flange.* The mounting flange shall be made of aluminum alloy conforming to Specification QQ–A–591 or QQ–A–601, or other material acceptable to the procuring activity. The mounting flange may be integral with the body of the switch.

3.3.4.1 *Threaded mounting inserts.* Unless otherwise specified, for mounting the switch, suitable No. 6-32NC-2B self-locking threaded inserts shall be provided. These inserts shall be of nonmagnetic, corrosion-resisting material and shall be securely and permanently attached to each mounting flange shown on the applicable AN standard.

3.3.5 *Operating mechanisms.* Any suitable mechanism which will operate without binding or excessive friction may be employed.

3.3.5.1 *Switch stationary contacts and movable contact segments.* Switch stationary contacts and movable contact segments shall be made of beryllium copper having Rockwell hardness values of not less than 15N–78 and not more than 15N–82.

3.3.6 *Terminals.* Terminals shall be of the screw or stud type with No. 8–32NC–2 threads.

3.3.7 *Conduit boss.* The conduit boss for the ignition switch shall be as specified on the applicable AN standard. The total resistance between the conduit boss and any grounded part of the switch mechanism shall exceed 0.0025 ohm.

3.3.8 *Interchangeability.* All parts having the same manufacturers part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of Specification MIL–D–70327.

3.3.9 *Weight.* The weight shall be as specified on the applicable AN standard.

3.3.10 *Dimensions.* The dimensions shall



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be as specified on the applicable AN standard.

**3.3.11** *Screw threads.* Screw threads shall conform to Specification MIL–S–7742.

**3.4 Clearances.**

**3.4.1** *Creepage and clearances.* The minimum creepage path between current-carrying parts and ground shall be 1/8 inch. The minimum clearance between current-carrying parts and other portions of the switch, other than insulating material, shall be a minimum of 1/16 inch.

**3.4.2** *Service clearance.* Sufficient clearance shall be provided between the switch body and the back of the can to facilitate installation wiring.

**3.5 Finish.**

**3.5.1** *Aluminum and aluminum-alloy parts.* Aluminum and aluminum-alloy parts, except switch handles and except all electrical joints (including shielding joints and threads for conduit connection parts), shall be protected by an anodic film conforming to Specification MIL–A–8625, or a zinc chromate primer in accordance with Specification MIL–P–8585. All threads for conduit connection parts shall be coated with petrolatum conforming to Specification VV–P–236.

**3.5.2** *Steel, brass, and other corrosive metal parts.* Steel, brass, and other corrosive metal parts shall be cadmium plated where practicable in accordance with Specification QQ–P–416, type II or III, as applicable, and of a class that is adequate to achieve the degree of protection required.

**3.5.3** *Exterior surfaces.* All exterior surfaces of the mounting flange, faceplate, and area behind the mounting flange, except threads, shall be painted with two coats of enamel conforming to type IV of Specification MIL–E–5557, baked after each coat.

**3.6 Performance.** When tested as specified in section 4, the switches shall conform to the following requirements.

**3.6.1** *Insulation.* When tested as specified in 4.6.2, there shall be no breakdown of insulation or damage.

**3.6.2** *Vibration.* When tested as specified in 4.6.3, there shall be no loosening or failure of parts.

**3.6.3** *Leakage.* When tested as specified in 4.6.4, there shall be no moisture on the interior surfaces of the switch.

**3.6.4** *Durability.* When tested as specified in 4.6.5, there shall be no mechanical or electrical failure, or impairment of the operating efficiency.

**3.6.5** *Cold resistance.* When tested as specified in 4.6.6, there shall be no mechanical or electrical damage to the switch.

**3.6.6** *Shock.* When tested as specified in 4.6.7, there shall be no opening of closed contacts or closing of open contacts. There shall be no mechanical or electrical damage to the switch.

**3.6.7** *Altitude.* When tested as specified in 4.6.8, there shall be no mechanical or electrical failure or damage which will affect the operation of the switch.

**3.6.8** *Humidity.* On completion of the drying period specified in 4.6.9, the switch shall be electrically and mechanically operative and the operating forces shall be within the limits specified in 4.6.14.

**3.6.9** *Sand and dust.* When tested as specified in 4.6.10, there shall be no mechanical or electrical damage which will prevent subsequent satisfactory operation of the switch.

**3.6.10** *Corrosion resistance.* When tested as specified in 4.6.11, there shall be no flak-



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ing of external finish, and no evidence of damage to the internal parts which would result in the inability of the switch to conform to any other performance requirements of this specification.

3.6.11 *Mounting nuts.* When tested as specified in 4.6.12, there shall be no loosening of the mounting nuts or damage to the switch.

3.6.12 *Mounting flange.* When tested as specified in 4.6.13, there shall be no fracture or damage.

3.6.13 *Operating forces.* When tested as specified in 4.6.14, the operating forces shall not exceed the limits specified.

3.6.14 *Preservation, packaging, packing, and marking.* When examined as specified in 4.7, preparation for delivery shall be in accordance with 5.1.

3.7 **Identification of product.** The switch shall have a nameplate securely attached to the body of the switch marked with the following information in accordance with Standard MIL-STD-130:

SWITCH, ROTARY, SHIELDED, AIRCRAFT IGNITION
AN part No.
Mfr's part No.
Mfr's serial No.
Stock No.
Contract or order No.
Mfr's name or trademark
U. S.

3.7.1 *Use of AN or MIL designations.* AN or MIL designations shall not be applied to a product, except for qualification inspection samples, nor referred to in correspondence or sales matter until notification has been received from the activity responsible for qualification that the product has been approved for listing on the applicable Qualified Products List.

3.7.2 *Faceplates.* The marking "IGNITION," "BOTH," "L," "R," and "OFF" shall be formed by a process producing raised or depressed letters showing bright metal and the letters shall conform to Standard MS33558. The AN part number, and the manufacturers name or trademark (omit address) shall be marked in black on black.

3.7.3 *Master switch.* The control knob of the master switch shall have depressed letters filled with black enamel.

3.8 **Installation.** An envelope, containing the mounting screws and installation instructions, and having the following information printed on its face, shall be furnished with each switch:

IMPORTANT
THIS ENVELOPE CONTAINS
INSTRUCTIONS
AND MOUNTING SCREWS

3.9 **Workmanship.** The switches, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges, accuracy of dimensions, thoroughness of soldering, and alignment of parts.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.



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4.2 **Classification of tests.** The inspection and testing of switches shall be classified as:

(a) Qualification tests (4.3)

(b) Quality conformance tests (4.4)

4.3 **Qualification tests.**

4.3.1 *Sampling instructions.* The qualification test samples shall consist of two switches of each manufacturer's part number upon which qualification is desired. Samples shall be identified as required and forwarded to the activity responsible for qualification designated in the letter of authorization from that activity (see 6.4).

4.3.2 *Prior qualification.* Unless otherwise specified by the procuring activity, switches which have not previously passed the qualification tests or which have passed the qualification tests and have been modified in any manner shall satisfactorily pass the qualification tests prior to the acceptance of any switch.

4.3.3 *Tests.* The qualification tests shall consist of all the tests as specified in 4.6 and they shall be performed in the following order:

| | |
|---|---|
| (a) Insulation | (4.6.2) |
| (b) Cold resistance | (4.6.6) |
| (c) Shock | (4.6.7) |
| (d) Vibration | (4.6.3) |
| (e) Altitude | (4.6.8) |
| (f) Humidity | (4.6.9) |
| (g) Leakage | (4.6.4) |
| (h) Durability | (4.6.5) |
| (i) Sand and dust | (4.6.10) |
| (j) Corrosion resistance | (4.6.11) |
| (k) Mounting nuts | (4.6.12) |
| (l) Mounting flange | (4.6.13) |
| (m) Operating forces | (4.6.14) |
| (n) Examination of product | (4.6.1) |

4.4 **Quality conformance tests.** The quality conformance tests shall consist of the individual tests and the sampling tests, 4.4.1 and 4.4.2, respectively.

4.4.1 *Individual tests.* Each switch shall be subjected to the test specified in 4.6.1, examination of product. In addition, each switch shall be subjected to any other test specified herein which the inspector considers necessary to determine compliance with this specification.

4.4.2 *Sampling tests.* The sampling tests shall consist of sompling plan A and sampling plan B tests.

4.4.2.1 *Sampling plan A.* The first two production switches on each order, and thereafter, sample switches as shown below, shall be subjected to the following tests as specified in 4.6:

| *Number of samples selected* | *Lot* |
|---|---|
| None | 1 to 10 |
| One item or 1 percent of the total quantity on order, whichever is the greater | 11 or more |

| | |
|---|---|
| (a) Insulation | (4.6.2) |
| (b) Vibration | (4.6.3) |
| (c) Leakage | (4.6.4) |
| (d) Durability (high and low temperature) | (4.6.5) |

4.4.2.2 *Sampling plan B.* Two switches selected at random, representing the manufacturer's contract or order, shall be subjected to the following tests in the order shown:

| | |
|---|---|
| (a) Cold resistance | (4.6.6) |
| (b) Shock | (4.6.7) |
| (c) Altitude | (4.6.8) |
| (d) Humidity | (4.6.9) |
| (e) Sand and dust | (4.6.10) |
| (f) Corrosion resistance | (4.6.11) |
| (g) Mounting nuts | (4.6.12) |
| (h) Mounting flange | (4.6.13) |
| (i) Operating forces | (4.6.14) |
| (j) Preservation, packaging, packing, and marking | (4.7) |



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**4.5 Test condtiions.**

4.5.1 *Temperature.* Unless otherwise specified, the switch shall be tested at an ambient temperature of 25° ±5° C.

4.5.2 *Pressure.* Unless otherwise specified, the pressure shall be approximately 30 inches Hg.

**4.6 Test methods.**

4.6.1 *Examination of product.* Each switch shall be inspected to determine compliance with the requirements of this specification with respect to materials, marking, and workmanship (see 3.2, 3.7, and 3.9).

4.6.2 *Insulation.* The switch shall be subjected to 550 volts root mean square (rms) commercial frequency between each terminal and ground for 60 seconds. For individual tests (see 4.4.1), the voltage shall be applied between the terminals collectively and ground.

4.6.3 *Vibration.* The ignition switch shall be vibration tested with the handle in the "BOTH" position. Vibration shall consist of simple harmonic motion, having an amplitude not exceeding 0.03 inch (maximum total excursion 0.06 inch) and a frequency varied uniformly between the approximate limits of 600 to 3,300 cycles per minute (cpm). The entire cycle (see 6.3) of frequencies shall be accomplished in approximately 3 to 5 minutes. The ignition switch shall be mounted with the longitudinal axis vertical or in either of two horizontal positions revolved 90 degrees. The position most likely to cause malfunctioning shall be selected and the ignition switch shall be vibrated continuously for a period of 7 hours. If mounting in more than one position is considered likely to cause malfunctioning, the ignition switch shall be vibrated continuously for 7 hours in each position. At the end of the test, the cover shall be removed and all parts inspected.

4.6.4 *Leakage.* The ignition switch shall be subjected to a water spray for 1 hour. The spray shall be so directed and applied to the switch as to simulate the conditions of a heavy rain. Switches shall not be immersed in water. During the test, the switch shall be revolved about its various axes in such manner that all parts of the outside surface will receive approximately the same amount of exposure to the spray of water. The conduit boss opening shall be sealed with a metal plate, with rubber gasket and conduit nut drawn up tightly, during this test. Upon the expiration of the specified spray-test period, the cover shall be immediately removed and the interior of the switch examined for the presence of any moisture on the interior surfaces of the switch.

4.6.5 *Durability (high and low temperature).* The switch shall be mounted with its axis in a horizontal position on a stand and connected for normal operation to a magneto in accordance with Specification MIL–E–25109, for a nine-cylinder engine. The magneto shall be operated at a shaft speed of 2,500 rpm. While under the electrical load from the magneto primary, the switch mechanism shall be operated through 5,000 complete switching operations (cycles). Each individual switch control handle and the master switch control plunger shall be operated between extreme points to break the primary circuit at a rate of 30 to 40 operations per minute. The first 2,500 operations shall be with the switch at a maximum temperature of –55° C and the second 2,500 operations shall be with the switch at a minimum temperature of +70° C. The switch shall be held at the test temperature a minimum of 2 hours immediately preceding the test. At the conclusion of the test, the cover plate shall be removed from the switch and the internal parts, particularly the contact surfaces, examined.

4.6.6 *Cold resistance.* The switch shall be subjected to a minimum temperature of –65° C for 48 hours.



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4.6.7 *Shock.* Switches shall be subjected to a transient decelerating force produced by securing the switch in each one of its rectangular positions in turn to sufficient mass, and dropping the assembly through such a height that when decelerated by resilient impact, a deceleration of 50g shall be obtained. The shock testing mechanism specified in Specification JAN–S–44, altered for switch mounting is suitable for conducting this test.

4.6.8 *Altitude.* The switch shall be operated at room temperature under the electrical conditions and at the rate specified for durability (high and low temperature, 4.6.5) for 1 minute. The pressure shall be approximately 3.4 inches Hg. absolute.

4.6.9 *Humidity.* The switch shall be subjected to humidity test Procedure I of Specification MIL–E–5272. After removal from the humidity chamber, the switch shall be subjected to 200 cycles of operation as specified for durability (high and low temperature, 4.6.5), except that standard conditions of temperature and pressure shall be maintained. The switch shall then be permitted to dry at room ambient for 72 hours. After the drying period, the switch shall be tested for operating forces (4.6.14) and shall perform within the limits specified.

4.6.10 *Sand and dust.* This test shall be conducted in accordance with sand and dust test Procedure I of Specification MIL–E–5272. During the test, switches shall be operated at 25° ±2° C for 500 cycles, and at 71° ±2° C for 500 cycles of making and breaking this rated load at the lowest rated dc voltage at a rate of 10 to 12 cycles per minute.

4.6.11 *Corrosion resistance.* The switch shall be subjected to 50 hours of salt spray Procedure I of Specification MIL–E–5272 Within 15 minutes after removal from the salt spray chamber, the switch shall be washed in running water not warmer than 37° C, accompanied by a slight brushing with a soft bristle brush to remove salt deposits. It shall then be dried 6 hours in a circulating air oven at approximately 57° C. After being permitted to remain at room ambient for 72 hours, the switch shall be subjected to the test for operating forces (4.6.14). Performance shall be within the limits specified

4.6.12 *Mounting nuts.* Each mounting nut shall have an axial load of 25 pounds applied to the nut in such manner as to tend to pull the nut out of the flange toward the rear of the case. A torque of 10 pound-inches shall be applied about the axis of the threads.

4.6 13 *Mounting flange.* The switch case, with mechanism removed, shall be mounted face downward on the movable head of a suitable testing machine with the face of the case in a horizontal plane so that the mounting flange receives no added support. A suitable pin shall be inserted through the hole in the mounting flange and attached to a pull strap. A load of 175 pounds shall be applied to each flange in a direction toward the front of the case for a period of 1 minute.

4.6.14 *Operating forces.* The forces required to operate the switch handles from each position to each adjacent position shall be checked to determine that they are within the limits specified in table I.

TABLE I. *Operating forces*

| Condition | Torque to operate handles (pound-inches) | Pull-off force when applicable [1] (pounds) |
|---|---|---|
| Initial ..... | 2½ to 5 | 10 to 18 |
| After humidity and corrosion. | 6 (max) | 20 (max) |

[1] Applicable for multiple-engine switches equipped with a pull-off handle.

**4.7 Preservation, packaging, packing, and marking.** Preparation for delivery shall be inspected for conformance to section 5.



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## 5. PREPARATION FOR DELIVERY

**5.1 Preservation, packaging, packing, and marking.** Switches shall be preserved and packaged in accordance with level A or C of Specification MIL–E–17555, as specified (see 6.2). They shall be packed in accordance with level A, B, or C, as specified (see 6.2), and marked in accordance with Specification MIL–E–17555.

## 6. NOTES

**6.1 Intended use.** The switches covered by this specification are intended for use in the ignition systems of one-, two-, and four-engine aircraft.

**6.2 Ordering data.** Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) AN part No. of the switch desired (see 1.2).
(c) Selection of applicable levels of preservation, packaging, and packing (see 5.1).

**6.3 Definitions.**

6.3.1 *Cycle.* A cycle consists of operation of the switch from one extreme position to the other extreme position and back to starting position.

**6.4 Qualification.** With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Bureau of Naval Weapons, Navy Department, Washington 25, D. C., and information pertaining to qualification of products may be obtained from that activity.

**Custodians:**
**Army—MO**
**Navy—Weps**
**Air Force—AFLC (SGEE)**

**Preparing activity:**
**Navy—Weps**
**Project No. 5930–0074**



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